

This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "[Guide to Using Scriptor Modules](#)" document.

Device Specifications

Device Type: Video Projector
Manufacturer: Christie
Firmware Version: N/A
Model(s): L2K1000, L2K1500

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scriptor Version
2.06.0000-b010	1.4.1

Version History

Module Version	Date	Notes
1_0_1_0	10/30/2017	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
Example: `InterfaceName.Unidirectional = 'True'`
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
Example: `InterfaceName.connectionCounter = 5`
- If login credentials are required, devicePassword must be set accordingly.
Example: `InterfaceName.devicePassword = 'extron'`

Supported Classes and Examples

SerialClass
<code>InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='L2K1000')</code>
SerialOverEthernetClass
<code>InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='L2K1000')</code>
EthernetClass
<code>InterfaceName = ModuleName.EthernetClass('192.168.254.254', 10000, Model='L2K1000')</code>

Set Commands

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command AspectRatio	Value 'Normal' 'Zoom' 'DZoom +'	Value 'Full' 'True' 'DZoom -'	Value 'Wide' 'Custom'
# AspectRatio example InterfaceName.Set('AspectRatio', 'Normal')			
Command AutoImage	Value None		
# AutoImage example InterfaceName.Set('AutoImage', None)			
Command ClosedCaption ¹	Value 'CC1' 'CC4'	Value 'CC2' 'Off'	Value 'CC3'
# ClosedCaption example InterfaceName.Set('ClosedCaption', 'CC1')			
Command Focus	Value 'Near'	Value 'Far'	
# Focus example InterfaceName.Set('Focus', 'Near')			
Command Freeze	Value 'On'	Value 'Off'	
# Freeze example InterfaceName.Set('Freeze', 'On')			
Command Input ²	Value 'Input 1 Digital' 'Input 1 SCART' 'Input 2 YPbPr' 'Input 3 Digital' 'Input 3 S-Video' 'Input 3 SCART' 'Input 3 HDMI' 'Input 4 Video'	Value 'Input 1 HDMI' 'Input 1 HDCP' 'Input 2 Video' 'Input 3 Analog' 'Input 3 SDI 1' 'Input 3 YPbPr' 'Input 4 Digital' 'Input 4 S-Video'	Value 'Input 1 Analog' 'Input 2 S-Video' 'Input 2 RGB' 'Input 3 Video' 'Input 3 SDI 2' 'Input 3 HDCP' 'Input 4 Analog' 'Input 4 SCART'

Global Scripter Module Communication Sheet

	'Input 4 YPbPr'	'Input 4 SDI 1'	'Input 4 SDI 2'
	'Input 4 HDCP'	'Input 4 HDMI'	
# Input example InterfaceName.Set('Input', 'Input 1 Digital')			
Command LampMode	Value 'Normal' 'Auto'	Value 'Eco 1'	Value 'Eco 2'
# LampMode example InterfaceName.Set('LampMode', 'Normal')			
Command LensShift	Value 'Up' 'Right'	Value 'Down'	Value 'Left'
# LensShift example InterfaceName.Set('LensShift', 'Up')			
Command MenuNavigation	Value 'Up' 'Right'	Value 'Down' 'Enter'	Value 'Left'
# MenuNavigation example InterfaceName.Set('MenuNavigation', 'Up')			
Command OnScreenDisplay	Value 'On'	Value 'Off'	
# OnScreenDisplay example InterfaceName.Set('OnScreenDisplay', 'On')			
Command Power	Value 'On'	Value 'Off'	
# Power example InterfaceName.Set('Power', 'On')			
Command VideoMute	Value 'On'	Value 'Off'	
# VideoMute example InterfaceName.Set('VideoMute', 'On')			
Command Zoom	Value 'In'	Value 'Out'	
# Zoom example InterfaceName.Set('Zoom', 'In')			

¹ Closed Caption is available when entered signal is NTSC from Video or S-Video input.

² Input 3 and 4 selection depends on the card that is inserted into the projector.

Status Available

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'}, FeedbackHandler)
FeedbackHandler will be called only when the specified qualifier gets a new status.
```

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

Command AspectRatio	Value 'Normal' 'Zoom'	Value 'Full' 'True'	Value 'Wide' 'Custom'
# AspectRatio examples InterfaceName.Update('AspectRatio') Value = InterfaceName.ReadStatus('AspectRatio') InterfaceName.SubscribeStatus('AspectRatio', None, FeedbackHandler)			
Command ClosedCaption ¹	Value 'CC1' 'CC4'	Value 'CC2' 'Off'	Value 'CC3'
# ClosedCaption examples InterfaceName.Update('ClosedCaption') Value = InterfaceName.ReadStatus('ClosedCaption') InterfaceName.SubscribeStatus('ClosedCaption', None, FeedbackHandler)			
Command ConnectionStatus	Value 'Connected'	Value 'Disconnected'	
# ConnectionStatus examples Value = InterfaceName.ReadStatus('ConnectionStatus') InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)			
Command DeviceStatus	Value 'Normal' 'Standby after Cooling Down (Abnormal Temp)' 'Process Cooling Down after Off (Lamp Failure)' 'Power Save'	Value 'Power Fail' 'Power Save (Cooling Down)' 'Process Cooling Down after Off (Shutter Mgmt)'	Value 'Process Cooling Down (Abnormal Temperature)' 'Standby after Cooling Down (Lamp Failure)' 'Standby after Cooling Down (Shutter Mgmt)'

# DeviceStatus examples InterfaceName.Update('DeviceStatus') Value = InterfaceName.ReadStatus('DeviceStatus') InterfaceName.SubscribeStatus('DeviceStatus', None, FeedbackHandler)			
Command FilterUsage	Value Hours		
# FilterUsage examples InterfaceName.Update('FilterUsage') Value = InterfaceName.ReadStatus('FilterUsage') InterfaceName.SubscribeStatus('FilterUsage', None, FeedbackHandler)			
Command Freeze	Value 'On'	Value 'Off'	
# Freeze examples InterfaceName.Update('Freeze') Value = InterfaceName.ReadStatus('Freeze') InterfaceName.SubscribeStatus('Freeze', None, FeedbackHandler)			
Command Input ²	Value 'Input 1 Digital'	Value 'Input 1 HDMI'	Value 'Input 1 Analog'
	'Input 1 SCART'	'Input 1 HDCP'	'Input 2 S-Video'
	'Input 2 YPbPr'	'Input 2 Video'	'Input 2 RGB'
	'Input 3 Digital'	'Input 3 Analog'	'Input 3 Video'
	'Input 3 S-Video'	'Input 3 SDI 1'	'Input 3 SDI 2'
	'Input 3 SCART'	'Input 3 YPbPr'	'Input 3 HDCP'
	'Input 3 HDMI'	'Input 4 Digital'	'Input 4 Analog'
	'Input 4 Video'	'Input 4 S-Video'	'Input 4 SCART'
	'Input 4 YPbPr'	'Input 4 SDI 1'	'Input 4 SDI 2'
	'Input 4 HDCP'	'Input 4 HDMI'	
# Input examples InterfaceName.Update('Input') Value = InterfaceName.ReadStatus('Input') InterfaceName.SubscribeStatus('Input', None, FeedbackHandler)			
Command LampMode	Value 'Normal'	Value 'Eco 1'	Value 'Eco 2'
	'Auto'		
# LampMode examples InterfaceName.Update('LampMode') Value = InterfaceName.ReadStatus('LampMode') InterfaceName.SubscribeStatus('LampMode', None, FeedbackHandler)			
Command LampStatus	Value 'On'	Value 'Off'	Value 'Failure'
Qualifier Key 'Lamp'	Qualifier Value '1' – '2'		

<pre># LampStatus examples InterfaceName.Update('LampStatus', {'Lamp': '1'}) Value = InterfaceName.ReadStatus('LampStatus', {'Lamp': '1'}) InterfaceName.SubscribeStatus('LampStatus', None, FeedbackHandler)</pre>			
Command LampUsage	Value Hours		
Qualifier Key 'Lamp'	Qualifier Value '1' – '2'		
<pre># LampUsage examples InterfaceName.Update('LampUsage', {'Lamp': '1'}) Value = InterfaceName.ReadStatus('LampUsage', {'Lamp': '1'}) InterfaceName.SubscribeStatus('LampUsage', None, FeedbackHandler)</pre>			
Command OperationHours	Value Hours		
<pre># OperationHours examples InterfaceName.Update('OperationHours') Value = InterfaceName.ReadStatus('OperationHours') InterfaceName.SubscribeStatus('OperationHours', None, FeedbackHandler)</pre>			
Command Power	Value 'On'	Value 'Off'	Value 'Warming Up'
	'Cooling Down'		
<pre># Power examples InterfaceName.Update('Power') Value = InterfaceName.ReadStatus('Power') InterfaceName.SubscribeStatus('Power', None, FeedbackHandler)</pre>			
Command VideoMute	Value 'On'	Value 'Off'	
<pre># VideoMute examples InterfaceName.Update('VideoMute') Value = InterfaceName.ReadStatus('VideoMute') InterfaceName.SubscribeStatus('VideoMute', None, FeedbackHandler)</pre>			

¹ Closed Caption is available when entered signal is NTSC from Video or S-Video input.² Input 3 and 4 selection depends on the card that is inserted into the projector.

Cable and Adapter Requirements

Custom Captive Screw to DB9 RS-232 Serial Cable

Notes for the Device

Serial communication

Port Type: RS-232

Baud Rate: 19200

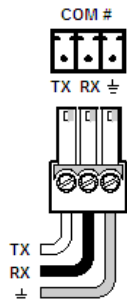
Data Bits: 8

Parity: None

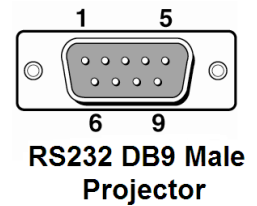
Stop Bits: One

Flow Control: None

Pin Assignments Diagram



Signal	Main Cable	Pin	Signal
TxD		2	TxD
RxD		3	RxD
GND		5	GND



Network communication

When configuring the Ethernet module, be sure device settings match those of the Global Scripter ethernet interface

Port Type:	Ethernet
Default Port:	10000
Logon Credentials Supported:	Yes
Default Password:	0000
Multi-Connection Capabilities:	Undetermined
Port Changeability:	Yes

Ethernet Module Configuration Description

Please refer to user manual for settings and changes to the network communication

Notes for the Device

Appendix A. Set Commands

Aspect Ratio Custom	CF SCREEN CUSTOM\x0D
Aspect Ratio DZoom -	CF SCREEN DZOOM DN\x0D
Aspect Ratio DZoom +	CF SCREEN DZOOM UP\x0D
Aspect Ratio Full	CF SCREEN FULL\x0D
Aspect Ratio Normal	CF SCREEN NORMAL\x0D
Aspect Ratio True	CF SCREEN TRUE\x0D
Aspect Ratio Wide	CF SCREEN WIDE\x0D
Aspect Ratio Zoom	CF SCREEN ZOOM\x0D
Auto Image None	CF KEYEMU AUTOPC\x0D
Closed Caption CC1	CF CCAPTIONDISP CC1\x0D
Closed Caption CC2	CF CCAPTIONDISP CC2\x0D
Closed Caption CC3	CF CCAPTIONDISP CC3\x0D
Closed Caption CC4	CF CCAPTIONDISP CC4\x0D
Closed Caption Off	CF CCAPTIONDISP OFF\x0D
Focus Far	C4B\x0D
Focus Near	C4A\x0D
Freeze Off	CF FREEZE OFF\x0D
Freeze On	CF FREEZE ON\x0D
Input Input 1 Analog	CF INPUT1 ANALOG\x0D
Input Input 1 Digital	CF INPUT1 DIGITAL\x0D
Input Input 1 HDCP	CF INPUT1 HDCP\x0D
Input Input 1 HDMI	CF INPUT1 HDMI\x0D
Input Input 1 SCART	CF INPUT1 SCART\x0D
Input Input 2 RGB	CF INPUT2 ANALOG\x0D
Input Input 2 S-Video	CF INPUT2 S-VIDEO\x0D
Input Input 2 Video	CF INPUT2 VIDEO\x0D
Input Input 2 YPbPr	CF INPUT2 YPBPR\x0D
Input Input 3 Analog	CF INPUT3 ANALOG\x0D
Input Input 3 Digital	CF INPUT3 DIGITAL\x0D
Input Input 3 HDCP	CF INPUT3 HDCP\x0D
Input Input 3 HDMI	CF INPUT3 HDMI\x0D
Input Input 3 SCART	CF INPUT3 SCART\x0D
Input Input 3 SDI 1	CF INPUT3 SDI1\x0D
Input Input 3 SDI 2	CF INPUT3 SDI2\x0D
Input Input 3 S-Video	CF INPUT3 S-VIDEO\x0D
Input Input 3 Video	CF INPUT3 VIDEO\x0D
Input Input 3 YPbPr	CF INPUT3 YPBPR\x0D
Input Input 4 Analog	CF INPUT4 ANALOG\x0D
Input Input 4 Digital	CF INPUT4 DIGITAL\x0D
Input Input 4 HDCP	CF INPUT4 HDCP\x0D
Input Input 4 HDMI	CF INPUT4 HDMI\x0D
Input Input 4 SCART	CF INPUT4 SCART\x0D

Input Input 4 SDI 1	CF INPUT4 SDI1\x0D
Input Input 4 SDI 2	CF INPUT4 SDI2\x0D
Input Input 4 S-Video	CF INPUT4 S-VIDEO\x0D
Input Input 4 Video	CF INPUT4 VIDEO\x0D
Input Input 4 YPbPr	CF INPUT4 YPBPR\x0D
Lamp Mode Auto	CF AUTOLAMPCTRL AUTO\x0D
Lamp Mode Eco 1	CF AUTOLAMPCTRL ECO1\x0D
Lamp Mode Eco 2	CF AUTOLAMPCTRL ECO2\x0D
Lamp Mode Normal	CF AUTOLAMPCTRL NORMAL\x0D
Lens Shift Down	C5E\x0D
Lens Shift Left	C5F\x0D
Lens Shift Right	C60\x0D
Lens Shift Up	C5D\x0D
Menu Navigation Down	CF KEYEMU DN\x0D
Menu Navigation Enter	CF KEYEMU SELECT\x0D
Menu Navigation Left	CF KEYEMU LEFT\x0D
Menu Navigation Right	CF KEYEMU RIGHT\x0D
Menu Navigation Up	CF KEYEMU UP\x0D
On Screen Display Off	CF MENU OFF\x0D
On Screen Display On	CF MENU ON\x0D
Power Off	CF POWER OFF\x0D
Power On	CF POWER ON\x0D
Video Mute Off	CF VMUTE OFF\x0D
Video Mute On	CF VMUTE ON\x0D
Zoom In	C47\x0D
Zoom Out	C46\x0D

Appendix B. Query Commands

Aspect Ratio	CR SCREEN\x0D
Closed Caption	CR CCAPTIONDISP\x0D
Device Status	CR STATUS\x0D
Filter Usage	CR FILH\x0D
Freeze	CR FREEZE\x0D
Input	CR INPUT\x0D
Lamp Mode	CR AUTOLAMPCTRL\x0D
Lamp Status Lamp 1	CR LAMPSTS\x0D
Lamp Status Lamp 2	CR LAMPSTS\x0D
Lamp Usage Lamp 1	CR LAMPH\x0D
Lamp Usage Lamp 2	CR LAMPH\x0D
Operation Hours	CR PROJH\x0D
Power	CR STATUS\x0D
Video Mute	CR VMUTE\x0D